

MELAKU ALEHEGN

Machine Learning | AI Engineer

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Summary

Software Engineer (AI/ML) Software Engineer with experience designing and deploying production-grade machine learning and LLM-based systems. Strong focus on end-to-end pipelines, data-intensive feature engineering, and integrating AI models into real-world products. Comfortable working across data, backend, and infrastructure layers to deliver reliable, scalable solutions.

Experience

Wicked Analytics

Remote

Machine Learning Engineer

02/2026 - Present

- Designed, trained, and deployed a 16-week ML demand forecasting model trained on multi-year historical Walmart sales and store-level data for retail-supplier clients, producing per-item weekly predictions that drive replenishment decisions.
- Built Airflow-orchestrated pipelines that pull daily sales, demand forecasts, supply plans, and supplier-side production data into Snowflake, transform them through DBT models, and distribute weekly outputs to client-facing storage and BI dashboards.
- Modeled the core data layer in DBT and Snowflake: per-client transformation logic for store-level sales, demand forecasts, supply plans, and supplier inventory.
- Built a full-stack demand planning web application (React, FastAPI, PostgreSQL) for planners: interactive item list, editable per-week order plans, projected-inventory cascade computed on the fly, and multi-vendor support, replacing legacy spreadsheet-only workflows.
- Integrated automated weekly Excel deliverables (forecasts, supply plans, valid-store reports) into existing client workflows with stable naming and consistent file locations, so planners can drop the outputs straight into established planning workbooks without disrupting their cadence.

Arifpay

Addis Ababa, Ethiopia

Data Scientist

01/2025 - 05/2026

- Built and deployed machine learning models for payment fraud detection, working with high-volume transactional data and severe class imbalance
- Designed behavioral and temporal feature pipelines to improve fraud signal quality, contributing to a 15% reduction in fraud incidents
- Developed a merchant credit scoring system using FastAPI, improving lending decision accuracy and increasing approval confidence by -40%
- Implemented continuous learning pipelines using MLflow and Celery to retrain and monitor models in production
- Built and maintained Airflow DAGS for ML workflows, enabling reliable scheduling and monitoring and helping prevent an estimated \$300K in potential annual losses
- Designed Metabase dashboards for executive-level KPIs (revenue, churn, acquisition, efficiency), enabling data-driven decision-making across the business
- Built Retrieval-Augmented Generation (RAG) systems over internal documents and operational data to support internal analytics and decision-support workflows, focusing on reliability and real-world usability

Openstack

Remote

Software Developer Intern

12/2023 - 04/2024

- Enhanced the UI workflow of the OpenStack Manila project using Django views, streamlining the share network creation process and improving usability.
- Increased test reliability by achieving 90% unit test coverage and introducing pre-commit and tox configurations to standardize development workflows.
- Collaborated with OpenStack core reviewers to develop, refine, and merge patches within release cycles in a large-scale open-source environment.

Nedamco Africa

Remote(Netherlands)

Cloud Consultant

12/2023 - 03/2024

- Consulted clients across agritech, hospitality, and pharmaceutical industries on cloud adoption strategies using AWS and Azure platforms.
- Assisted in designing scalable cloud architectures and guided clients on selecting services for compute, storage, networking, monitoring, and cost optimization.
- Supported cloud migration and deployment planning while working with AWS services including EC2, S3, IAM, RDS, and CloudWatch.
- Participated in stakeholder discussions and prepared technical recommendations, architecture documentation, and implementation guidance aligned with business requirements.

Experience

Mmcy tech

Addis Ababa, Ethiopia

Full stack developer intern

07/2023 - 01/2024

- Developed a full-stack Applicant Tracking System (ATS) web app using Vue.js, Express.js, and Tailwind CSS, designing schemas and API endpoints for user authentication and job management
- Implemented core ATS features including login, candidate listings, job listings, and an admin dashboard, enhancing recruiter productivity with streamlined workflows
- Collaborated in an agile environment to build and maintain software applications and databases, ensuring timely delivery of project milestones

Education

Addis Ababa Science & Technology University

Addis Ababa, Ethiopia

BSc in Electrical and Computer Engineering

09/2016 - 09/2021

Skills

Applied AI / ML: Python • Pandas • NumPy • Scikit-learn • TensorFlow

Data & MLOps: Apache Airflow • MLflow • Celery • Kafka • Continuous Training • Data Pipelines

Backend & Deployment: FastAPI • Flask • Docker • REST APIs

LLMs & Search: LangChain • Vector Databases (ChromaDB)

Cloud & Analytics: AWS • GCP • Metabase • PowerBI • Azure